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4.5. Earthquake Damage in Kavrepalanchok NP

In this assignment, you'll build a classification model to predict building damage for the district of Kavrepalanchok.

```
[54]: import warnings
import sqlite3
warnings.simplefilter(action="ignore", category=FutureWarning)
warnings.resetwarnings()

[55]: # Import libraries here
import sqlite3
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from category_encoders import OneHotEncoder
from category_encoders import OrdinalEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline, make_pipeline
from sklearn.tree import DecisionTreeClassifier, plot_tree
```

Prepare Data

Connect

Run the cell below to connect to the `nepal.sqlite` database.

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Prepare Data

Connect

Run the cell below to connect to the `nepal.sqlite` database.

```
[56]: %load_ext sql
%sql sqlite:///home/jovyan/nepal.sqlite

The sql extension is already loaded. To reload it, use:
%reload_ext sql

[56]: 'Connected: @/home/jovyan/nepal.sqlite'
```

Warning: Be careful with your SQL queries in this assignment. If you try to get all the rows from a table (for example, `SELECT * FROM id_map`), you will cause an Out of Memory error on your virtual machine. So always include a `LIMIT` when first exploring a database.

Task 4.5.1: What districts are represented in the `id_map` table? Determine the unique values in the `district_id` column.

```
[58]: %sql
SELECT name
FROM sqlite_schema
WHERE type = "table"

+ sqlite:///home/jovyan/nepal.sqlite
Done.
```

name
id_map
building_structure
building_damage
household_demographics

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041-sqlite.ipynb x 044-demographics.ipynb x 045-assignment.ipynb

Python 3 (pykernel)

```

run sqlite_schema
WHERE type = "table"

* sqlite:///home/jovyan/nepal.sqlite
Done.

[50]:
    name
    id_map
    building_structure
    building_damage
    household_demographics

[82]: %sql
SELECT distinct(district_id)
FROM id_map

* sqlite:///home/jovyan/nepal.sqlite
Done.

[82]: district_id
1
2
3
4

[83]: result = _Dataframe().squeeze() # noqa F821
wqet_grader.grade("Project 4 Assessment", "Task 4.5.1", result)

```

Python master Score: 1

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Python 3 (pykernel)

```

wqet_grader.grade("Project 4 Assessment", "Task 4.5.1", result)

Python master Score: 1

What's the district ID for Kavrepalanchok? From the lessons, you already know that Gorkha is 4; from the textbook, you know that Ramechhap is 2. Of the remaining districts, Kavrepalanchok is the one with the largest number of observations in the id_map table.

Task 4.5.2: Calculate the number of observations in the id_map table associated with district 1.

[84]: %sql
SELECT count(*)
FROM id_map
WHERE district_id = 1

* sqlite:///home/jovyan/nepal.sqlite
Done.

[84]: count(*)
36112

[85]: result = [_Dataframe().astype(float).squeeze()] # noqa F821
wqet_grader.grade("Project 4 Assessment", "Task 4.5.2", result)

Yes! Your hard work is paying off.
Score: 1

Task 4.5.3: Calculate the number of observations in the id_map table associated with district 3.

[ ]: %sql

```

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Python 3 (pykernel)

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Import

Task 4.5.5: Write a wrangle function that will use the query you created in the previous task to create a DataFrame. In addition your function should:

1. Create a "severe_damage" column, where all buildings with a damage grade greater than 3 should be encoded as 1. All other buildings should be encoded as 0.

2. Drop any columns that could cause issues with leakage or multicollinearity in your model.

[92]:

```
# Build your "wrangle" function here
def wrangle(db_path):
    # Connect to database using connect method
    conn = sqlite3.connect(db_path)

    # Construct query
    query = """
    SELECT distinct(i.building_id) AS b_id,
           s.*,
           d.damage_grade
    FROM id_map AS i
    JOIN building_structure AS s ON i.building_id = s.building_id
    JOIN building_damage AS d ON i.building_id = d.building_id
    WHERE district_id = 3
    """

    # Read query results into DataFrame
    df = pd.read_sql(query, conn, index_col="b_id")

    # Identify leaky columns
    drop_cols = [col for col in df.columns if "post_eq" in col]

    # Create binary target
    df["damage_grade"] = df["damage_grade"].str[-1].astype(int)
    df["severe_damage"] = (df["damage_grade"] > 3).astype(int) # encode as 0's and 1's

    # Drop old target
    drop_cols.append("damage_grade")

    # Drop multicollinearity column
    drop_cols.append("count_floors_pre_eq")
```

SimplePython 3 (pykernel) | Busy

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35°CSmoke

Search

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Applied Data Science LabMy Path Module | WorldQuantUwork/ds-curricu (3) - JupyterLabData-Science-Lab(6) wrangle0.py

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046-data-d...a month ago

```
# Drop high categorical features
drop_cols.append("building_id")

# Drop columns
df.drop(columns=drop_cols, inplace=True)

return df
```

Use your wrangle function to query the database at "/home/jovyan/nepal.sqlite" and return your cleaned results.

[93]:

```
df = wrangle("/home/jovyan/nepal.sqlite")
df.head()
```

[93]:

b_id	age_building	plinth_area_sq_ft	height_ft_pre_eq	land_surface_condition	foundation_type	roof_type	ground_floor_type	other_floor_type	position	plan_configuration	superstructure	severe_damage
87473	15	382	18	Flat	Mud mortar-Stone/Brick	Bamboo/Timber-Light roof	Mud	Timber/Bamboo-Mud	Not attached	Rectangular	Stone, mud mortar	
87479	12	328	7	Flat	Mud mortar-Stone/Brick	Bamboo/Timber-Light roof	Mud	Not applicable	Not attached	Rectangular	Stone, mud mortar	
87482	23	427	20	Flat	Mud mortar-Stone/Brick	Bamboo/Timber-Light roof	Mud	Timber/Bamboo-Mud	Not attached	Rectangular	Stone, mud mortar	
87491	12	427	14	Flat	Mud mortar-Stone/Brick	Bamboo/Timber-Light roof	Mud	Timber/Bamboo-Mud	Not attached	Rectangular	Stone, mud mortar	
87496	32	360	18	Flat	Mud mortar-Stone/Brick	Bamboo/Timber-Light roof	Mud	Timber/Bamboo-Mud	Not attached	Rectangular	Stone, mud mortar	

[94]:

```
wget_grader.grade(
    "Project 4 Assessment", "Task 4.5.5", wrangle("/home/jovyan/nepal.sqlite")
)
```

Yup. You got it.

Score: 1

SimplePython 3 (pykernel) | Idle

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35°CSmoke

Search

ENG12:2613-03-2023

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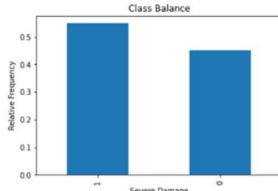
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Explore

Task 4.5.6: Are the classes in this dataset balanced? Create a bar chart with the normalized value counts from the "severe_damage" column. Be sure to label the x-axis "Severe Damage" and the y-axis "Relative Frequency". Use the title "Kavrepalanchok, Class Balance".

```
[95]: # Plot value counts of "severe_damage"
df["severe_damage"].value_counts(normalize=True).plot(
    kind="bar", xlabel="Severe Damage", ylabel="Relative Frequency", title="Class Balance"
);
# Don't delete the code below
plt.savefig("Images/4-5-6.png", dpi=150)
```



```
[96]: with open("Images/4-5-6.png", "rb") as file:
wqet_grader.grade("Project 4 Assessment", "Task 4.5.6", file)
```

Excellent work.
Score: 1

Task 4.5.7: Is there a relationship between the footprint size of a building and the damage it sustained in the earthquake? Use seaborn to create a boxplot that shows the distributions of the "plinth_area_sq_ft" column for both groups in the "severe_damage" column. Label your x-axis "Severe Damage" and y-axis "Plinth Area [sq. ft.]". Use the title "Kavrepalanchok, Plinth Area vs Building Damage".

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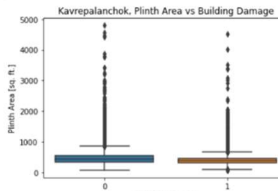
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Task 4.5.7: Is there a relationship between the footprint size of a building and the damage it sustained in the earthquake? Use seaborn to create a boxplot that shows the distributions of the "plinth_area_sq_ft" column for both groups in the "severe_damage" column. Label your x-axis "Severe Damage" and y-axis "Plinth Area [sq. ft.]". Use the title "Kavrepalanchok, Plinth Area vs Building Damage".

```
[97]: sns.boxplot(x="severe_damage", y="plinth_area_sq_ft", data=df)
plt.xlabel("Severe Damage")
plt.ylabel("Plinth Area [sq. ft.]")
plt.title("Kavrepalanchok, Plinth Area vs Building Damage");
# Don't delete the code below
plt.savefig("Images/4-5-7.png", dpi=150)
```



```
[98]: with open("Images/4-5-7.png", "rb") as file:
wqet_grader.grade("Project 4 Assessment", "Task 4.5.7", file)
```

Python master 🏆
Score: 1

Task 4.5.8: Are buildings with certain roof types more likely to suffer severe damage? Create a pivot table of 'df' where the index is "roof_type" and the values come from the "severe_damage" column, aggregated by the mean.

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Python 3 (pykernel)

wget_grader.grade("Project 4 Assessment", "Task 4.5.7", #file)

Python master

Score: 1

Task 4.5.8: Are buildings with certain roof types more likely to suffer severe damage? Create a pivot table of df where the index is "roof_type" and the values come from the "severe_damage" column, aggregated by the mean.

[99]:

```
roof_pivot = pd.pivot_table(df, index="roof_type", values="severe_damage", aggfunc=np.mean # roof_type: column in table).sort_values(by="severe_damage")
roof_pivot
```

[99]:

roof_type	severe_damage
RCC/RB/RBC	0.040715
Bamboo/Timber-Heavy roof	0.569477
Bamboo/Timber-Light roof	0.604842

[100]:

```
wget_grader.grade("Project 4 Assessment", "Task 4.5.8", roof_pivot)
```

Awesome work.

Score: 1

Split

Task 4.5.9: Create your feature matrix X and target vector y. Your target is "severe_damage".

Python 3 (pykernel)

wget_grader.grade("Project 4 Assessment", "Task 4.5.8", roof_pivot)

Awesome work.

Score: 1

Split

Task 4.5.9: Create your feature matrix X and target vector y. Your target is "severe_damage".

[101]:

```
X = df.drop(columns="severe_damage") # feature matrix: all columns apart from severe_damage
y = df["severe_damage"] # target vector
print("X shape:", X.shape)
print("y shape:", y.shape)
X shape: (76533, 11)
y shape: (76533,)
```

[102]:

```
wget_grader.grade("Project 4 Assessment", "Task 4.5.9a", X)
```

Yup. You got it.

Score: 1

[104]:

```
wget_grader.grade("Project 4 Assessment", "Task 4.5.9b", y)
```

Python master

Score: 1

Task 4.5.10: Divide your dataset into training and validation sets using a randomized split. Your validation set should be 20% of your data.

[]:

```
X_train, X_val, y_train, y_val = ...
print("X_train shape:", X_train.shape)
```

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Python master
Score: 1

Task 4.5.10: Divide your dataset into training and validation sets using a randomized split. Your validation set should be 20% of your data.

```
[105]: X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2, random_state=42)
print("X_train shape:", X_train.shape)
print("y_train shape:", y_train.shape)
print("X_val shape:", X_val.shape)
print("y_val shape:", y_val.shape)

X_train shape: (61226, 11)
y_train shape: (61226,)
X_val shape: (15307, 11)
y_val shape: (15307,)
```

uqet_grader.grade("Project 4 Assessment", "Task 4.5.10", [X_train.shape == (61226, 11)])

You got it. Dance party time!
Score: 1

Build Model

Baseline

Task 4.5.11: Calculate the baseline accuracy score for your model.

```
[106]: acc_baseline = y_train.value_counts(normalize=True).max() # normalize gives you the relative freq
print("Baseline Accuracy:", round(acc_baseline, 2))

Baseline Accuracy: 0.55

uqet_grader.grade("Project 4 Assessment", "Task 4.5.11", [acc_baseline])
```

Very impressive.
Score: 1

Iterate

Task 4.5.12: Create a model `model_lr` that uses logistic regression to predict building damage. Be sure to include an appropriate encoder for categorical features.

```
[110]: model_lr = make_pipeline(
    OneHotEncoder(use_cat_names=True),
    LogisticRegression(max_iter=1000-3000) #max_iter varies: suppresses the "ConvergenceWarning"
```

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Iterate

Task 4.5.12: Create a model `model_lr` that uses logistic regression to predict building damage. Be sure to include an appropriate encoder for categorical features.

```
[50]: model_lr = make_pipeline(
    OneHotEncoder(use_cat_names=True),
    LogisticRegression(max_iter=1000) # max_iter varies: suppresses the 'ConvergenceWarning'
)
# Fit model to training data
model_lr.fit(X_train, y_train)

[51]: wqet_grader.grade("Project 4 Assessment", "Task 4.5.12", model_lr)
```

You are coding
Score: 1

Task 4.5.13: Calculate training and validation accuracy score for `model_lr`.

```
[ ]: lr_train_acc = accuracy_score(y_train, model_lr.predict(X_train))
lr_val_acc = model_lr.score(X_val, y_val)

print("Logistic Regression, Training Accuracy Score:", lr_train_acc)
print("Logistic Regression, Validation Accuracy Score:", lr_val_acc)
```

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```
[52]: lr_train_acc = accuracy_score(y_train, model_lr.predict(X_train))
lr_val_acc = model_lr.score(X_val, y_val)

print("Logistic Regression, Training Accuracy Score:", lr_train_acc)
print("Logistic Regression, Validation Accuracy Score:", lr_val_acc)

Logistic Regression, Training Accuracy Score: 0.6513735994512135
Logistic Regression, Validation Accuracy Score: 0.6530998889397008

[53]: submission = [lr_train_acc, lr_val_acc]
wqet_grader.grade("Project 4 Assessment", "Task 4.5.13", submission)
```

Correct.
Score: 1

Task 4.5.14: Perhaps a decision tree model will perform better than logistic regression, but what's the best hyperparameter value for `max_depth`? Create a `for` loop to train and evaluate the model `model_dt` at all depths from 1 to 15. Be sure to use an appropriate encoder for your model, and to record its training and validation accuracy scores at every depth. The grader will evaluate your validation accuracy scores only.

```
[*]: depth_hyperparams = range(1, 16) # for max_depth
training_acc = []
validation_acc = []
for d in depth_hyperparams:
    model_dt = make_pipeline(
        OrdinalEncoder(),
        DecisionTreeClassifier(max_depth=d, random_state=42)
    )
    # Fit model to training data
    model_dt.fit(X_train, y_train)
    # Calculate training accuracy score and append to 'training_acc'
    training_acc.append(model_dt.score(X_train, y_train))
    # Calculate validation accuracy score and append to 'validation_acc'
    validation_acc.append(model_dt.score(X_val, y_val))

print("Training Accuracy Scores:", training_acc[1:15])
```

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36°C Smoke

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045-assignment.ipynb

Python 3 (pykernel)

Task 4.5.14: Perhaps a decision tree model will perform better than logistic regression, but what's the best hyperparameter value for `max_depth`? Create a `for` loop to train and evaluate the model `model_dt` at all depths from 1 to 15. Be sure to use an appropriate encoder for your model, and to record its training and validation accuracy scores at every depth. The grader will evaluate your validation accuracy scores only.

```
[54]: depth_hyperparams = range(1, 16) # for max_depth
training_acc = []
validation_acc = []
for d in depth_hyperparams:
    model_dt = make_pipeline(
        OrdinalEncoder(),
        DecisionTreeClassifier(max_depth=d, random_state=42)
    )
    # Fit model to training data
    model_dt.fit(X_train, y_train)
    # Calculate training accuracy score and append to 'training_acc'
    training_acc.append(model_dt.score(X_train, y_train))
    # Calculate validation accuracy score and append to 'validation_acc'
    validation_acc.append(model_dt.score(X_val, y_val))

print("Training Accuracy Scores:", training_acc[6])
print("Validation Accuracy Scores:", validation_acc[6])

Training Accuracy Scores: [0.6303041191650606, 0.6303041191650606, 0.642292490118577, 0.653529546271192, 0.6543951915852743, 0.6576617776761506]
Validation Accuracy Scores: [0.6350035931273273, 0.6350035931273273, 0.6453909975828053, 0.6527732410080493, 0.6529039001763899, 0.6504569151368654]

[55]: submission = pd.Series(validation_acc, index=depth_hyperparams)
wqet_grader.grade("Project 4 Assessment", "Task 4.5.14", submission)
```

✓

That's the right answer. Keep it up!
Score: 1

Task 4.5.15: Using the values in `training_acc` and `validation_acc`, plot the validation curve for `model_dt`. Label your x-axis "Max Depth" and your y-axis "Accuracy Score". Use the title "Validation Curve, Decision Tree Model", and include a legend.

Simple Python 3 (pykernel) | Idle

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Python 3 (pykernel)

[56]: # Validation curve
plt.plot(depth_hyperparams, training_acc, label="Training")
plt.plot(depth_hyperparams, validation_acc, label="validation")
plt.xlabel("Max Depth")
plt.ylabel("Accuracy Score")
plt.title("Validation Curve, Decision Tree Model")
plt.legend();

build & fit again
final_model_dt = make_pipeline(
 OrdinalEncoder(),
 DecisionTreeClassifier(max_depth=10, random_state=42)
)

Don't delete the code below
plt.savefig("Images/4-5-15.png", dpi=150)

[57]: with open("Images/4-5-15.png", "rb") as file:
wqet_grader.grade("Project 4 Assessment", "Task 4.5.15", file)

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Python 3 (pykernel)

✓ You got it. Dance party time! 🎉
Score: 1

Task 4.5.16: Build and train a new decision tree model `final_model_dt`, using the value for `max_depth` that yielded the best validation accuracy score in your plot above.

```
[58]: # Fit model to training data
final_model_dt.fit(X, y) #final_model_dt.fit(X_train, y_train)
```

✓ Pipeline

- OrdinalEncoder
- DecisionTreeClassifier

```
[59]: wqet_grader.grade("Project 4 Assessment", "Task 4.5.16", final_model_dt)
```

✓ Yes! Keep on rockin'. 🎸 That's right.
Score: 1

Evaluate

Task 4.5.17: How does your model perform on the test set? First, read the CSV file `"data/kavrepalanchok-test-features.csv"` into the DataFrame `X_test`. Next, use `final_model_dt` to generate a list of test predictions `y_test_pred`. Finally, submit your test predictions to the grader to see how your model performs.

Tip: Make sure the order of the columns in `X_test` is the same as in your `X_train`. Otherwise, it could hurt your model's performance.

```
[61]: X_test = pd.read_csv("data/kavrepalanchok-test-features.csv", index_col="b_id")
y_test_pred = pd.Series(final_model_dt.predict(X_test))
y_test_pred[:5]
```

Simple Python 3 (pykernel) | Idle

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Python 3 (pykernel)

Evaluate

Task 4.5.17: How does your model perform on the test set? First, read the CSV file `"data/kavrepalanchok-test-features.csv"` into the DataFrame `X_test`. Next, use `final_model_dt` to generate a list of test predictions `y_test_pred`. Finally, submit your test predictions to the grader to see how your model performs.

Tip: Make sure the order of the columns in `X_test` is the same as in your `X_train`. Otherwise, it could hurt your model's performance.

```
[94]: X_test = pd.read_csv("data/kavrepalanchok-test-features.csv", index_col="b_id")
y_test_pred = pd.Series(final_model_dt.predict(X_test))
y_test_pred[:5]
```

```
[94]: 0 1
1 1
2 0
3 1
4 0
dtype: int64
```

```
[96]: test_acc = final_model_dt.score(X_test, y_test_pred)
print("Test Accuracy:", round(test_acc, 2))
Test Accuracy: 1.0
```

```
[98]: acc_train = accuracy_score(y_train, model_lr.predict(X_train))
acc_test = model_lr.score(X_test, y_test_pred)
print("LR Training Accuracy:", acc_train)
print("LR Validation Accuracy:", acc_test)
```

LR Training Accuracy: 0.6513735994512135
LR Validation Accuracy: 0.8438073862477046

```
[99]: submission = pd.Series(y_test_pred)
wqet_grader.grade("Project 4 Assessment", "Task 4.5.17", submission)
```

Simple Python 3 (pykernel) | Idle

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Communicate Results

Task 4.5.18: What are the most important features for `final_model_dt`? Create a Series `Gini_feat_imp`, where the index labels are the feature names for your dataset and the values are the feature importances for your model. Be sure that the Series is sorted from smallest to largest feature importance.

```
[100]: features = X_train.columns
importances = final_model_dt.named_steps["decisiontreeclassifier"].feature_importances_
feat_imp = pd.Series(importances, index=features).sort_values()
feat_imp.head()
```

```
[100]: plan_configuration      0.004032
position                    0.007129
land_surface_condition     0.008241
ground_floor_type          0.009741
foundation_type            0.010620
dtype: float64
```

```
[101]: features = model_lr.named_steps["onehotencoder"].get_feature_names()
importances = model_lr.named_steps["logisticregression"].coef_[0]
feat_imp = pd.Series(np.exp(importances), index=features).sort_values()
feat_imp.head()
```

```
[101]: foundation_type_RC      0.430515
superstructure_Brick, cement mortar  0.550657
roof_type_RC/RB/RBC       0.560908
ground_floor_type_RC      0.599185
superstructure_RC, non-engineered    0.616918
dtype: float64
```

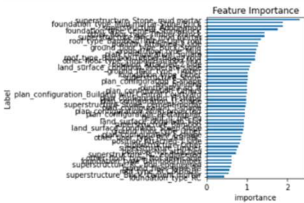
```
[102]: wqet_grader.grade("Project 4 Assessment", "Task 4.5.18", feat_imp)
```

Exception: Traceback (most recent call last)

Task 4.5.19: Create a horizontal bar chart of `feat_imp`. Label your x-axis "Gini Importance" and your y-axis "Label". Use the title "Kavrepalanchok Decision Tree, Feature Importance".

Do you see any relationship between this plot and the exploratory data analysis you did regarding roof type?

```
[103]: # Create horizontal bar chart of feature importances
# Horizontal bar chart
feat_imp.plot(kind='barh')
plt.xlabel("Importance")
plt.ylabel("Label")
plt.title("Feature Importance");
# Don't delete the code below
plt.tight_layout()
plt.savefig("images/4-5-19.png", dpi=150)
```



```
[104]: with open("images/4-5-19.png", "rb") as file:
wqet_grader.grade("Project 4 Assessment", "Task 4.5.19", file)
```